

Math 241, F14
Quiz 7

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Use the Chain Rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ when $z = e^{x+2y}$ with $x = s/t$ and $y = t/s$.

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s} \\ &= e^{x+2y} \cdot \frac{1}{t} + 2e^{x+2y} \cdot \left(-\frac{t}{s^2}\right) \\ &= e^{x+2y} \left(\frac{1}{t} - \frac{2t}{s^2}\right) = e^{\left(\frac{s}{t} + \frac{2t}{s}\right)} \left(\frac{1}{t} - \frac{2t}{s^2}\right)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t} \\ &= e^{x+2y} \left(-\frac{s}{t^2}\right) + 2e^{x+2y} \left(\frac{1}{s}\right) \\ &= e^{x+2y} \left(\frac{2}{s} - \frac{s}{t^2}\right) = e^{\left(\frac{s}{t} + \frac{2t}{s}\right)} \left(\frac{2}{s} - \frac{s}{t^2}\right)\end{aligned}$$

Problem 2. (5 points) Use the Chain Rule to compute $\frac{\partial R}{\partial x}$ when $x = y = 1$, where

$$R = \ln(u^2 + v^2 + w^2), \quad u = x + 2y, \quad v = 2x - y, \quad w = 2xy.$$

$$\begin{aligned}\frac{\partial R}{\partial x} &= \frac{\partial R}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial R}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial R}{\partial w} \frac{\partial w}{\partial x} \\ &= \frac{2u}{u^2+v^2+w^2} (1) + \frac{2v}{u^2+v^2+w^2} (2) + \frac{2w}{u^2+v^2+w^2} (2y) \\ &= \frac{2u+4v+4wy}{u^2+v^2+w^2}\end{aligned}$$

$$x=y=1 \Rightarrow u=3, v=1, w=2$$

$$\Rightarrow \frac{\partial R}{\partial x} \Big|_{x=y=1} = \frac{2 \cdot 3 + 4 \cdot 1 + 4 \cdot 2 \cdot 1}{3^2 + 1^2 + 2^2} = \frac{18}{14} = \frac{9}{7}$$